

Clinical Digital Twin Report

Computational Cellular Reprogramming Analysis

Session ID: session_2026-01-01T10-44-34-029Z

Run ID: #2

Date Generated: 2026-01-01 10:44:38

Total Cells: 2000 | Model Mode: CLINICAL

1. Introduction

Cellular reprogramming through the induction of pluripotency is a pivotal area in regenerative medicine, with significant implications for disease modeling and therapeutic applications. This experiment aimed to evaluate the efficacy of the Yamanaka factors (OSKM) in reprogramming somatic cells under conditions of oncogenic stress.

2. Materials and Methods

The computational approach involved consecutive OSKM transfections under oncogenic stress to induce cellular reprogramming. Despite the planned use of scVI for single-cell transcriptomic analysis, no such analyses were conducted in this run.

Experimental Interventions:

- Oncogenic Stress Induced
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)

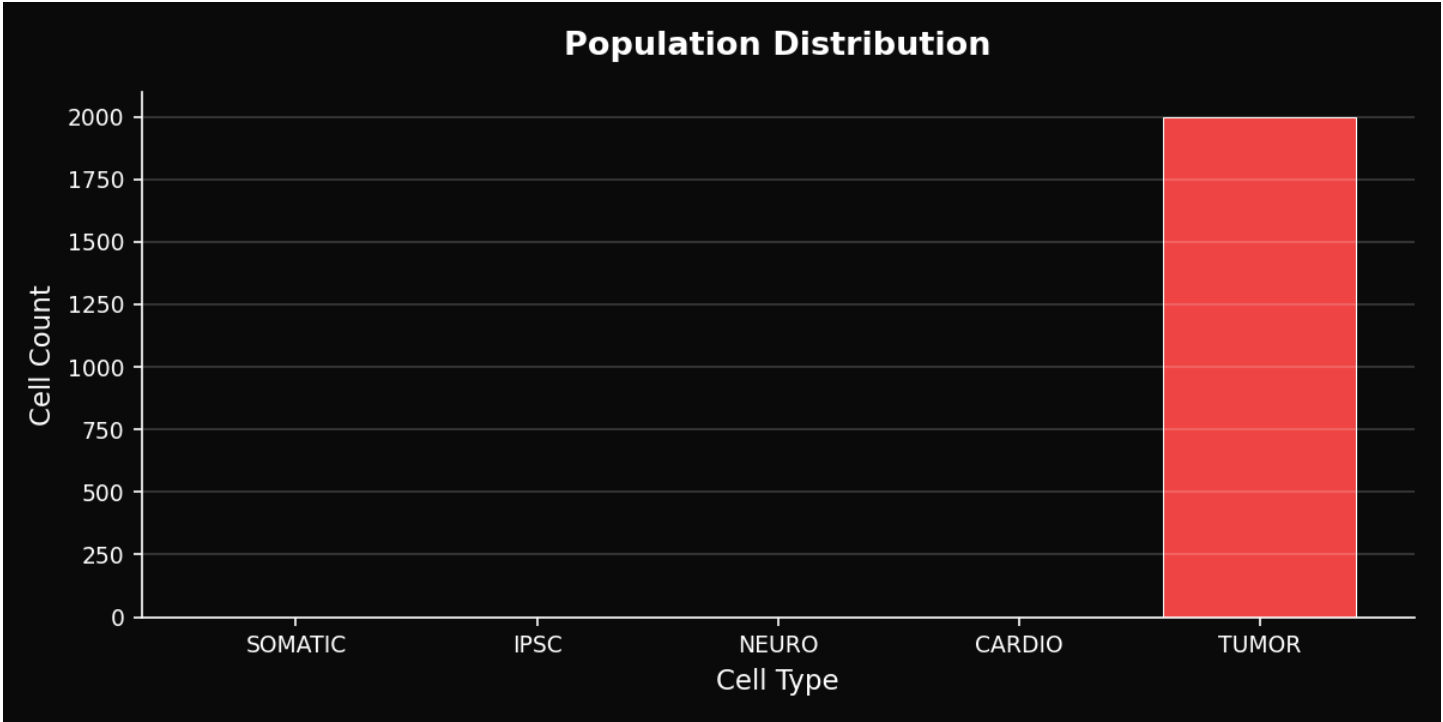
3. Results

The final cellular population analysis revealed a complete absence of somatic, induced pluripotent stem cells (iPSCs), neurogenic, and cardiogenic lineages, with a predominant presence of tumor cells amounting to 2000. This outcome suggests a pronounced shift towards tumorigenesis under the imposed experimental conditions.

3.1 Population Distribution

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Cell Type	Count	Percentage
SOMATIC	0	0.0%
IPSC	0	0.0%
NEURO	0	0.0%
CARDIO	0	0.0%
TUMOR	2000	100.0%

4. Discussion

The lack of iPSC generation and the overwhelming emergence of tumor cells indicate that the oncogenic stress, coupled with repeated OSKM transfections, skewed cellular fate towards tumorigenesis. This underscores the potential risks associated with reprogramming in oncogenic environments and highlights the need for careful modulation of reprogramming factors and conditions to prevent oncogenic transformations. Future investigations should explore alternative strategies to mitigate these effects and examine the underlying mechanisms driving tumorigenesis in this context.

This report was automatically generated by the IS-CHRP v25.0 Clinical AI Bridge platform. The data represents computational simulation results from a high-fidelity cellular reprogramming model.